

Ques 12)

Draw NFA- $\lambda$  for  $r = (0)11^* + (101)^*0$  & then convert the equivalent NFA & FA in the same

→

$$r = (0)11^* + (101)^*0$$

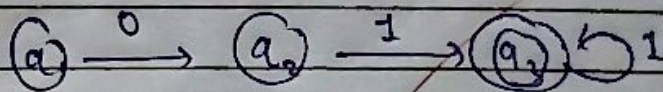
i)  $(0)11^*$ :

String starting with 0 followed by one or more 1's  
e.g.: 01, 011, 0111, 01111, ...

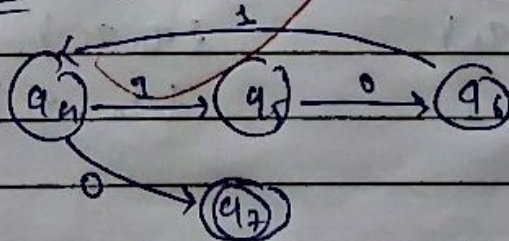
ii)  $(101)^*0$ :

Zero or more repetition of 101 followed by 0  
e.g.: 0, 1010, 10101, 101010, ...

Step 1: NFA- $\lambda$  for  $(0)11^*$



Step 2: NFA- $\lambda$  for  $(101)^*0$



Step 3: Combine both using  $\lambda$  transition

$$(0)11^* + (101)^*0$$

